Assingment 3
ADA 2601, 2602, 2603 - M
2026-2027 y.

Part I. Mann–Whitney, U - Test

Task 1. Comparing Student Performance Under Two Teaching Methods.

A university researcher wants to determine whether students taught using an interactive teaching method perform differently from students taught using a traditional teaching method. The examination scores of two independent groups are recorded below.

Interactive method	78	85	90	74	88	92
Traditional method	65	72	68	60	75	70

At the 5% significance level, determine whether there is a statistically significant difference between the two groups.

Task 2. Comparing Customer Satisfaction Between Two Banks.

A market researcher compares customer satisfaction ratings from two independent banks. Customers rate their satisfaction on a scale from 1 to 10.

Bank A	8	9	7	10	8	9	7
Bank B	7	6	8	5	7	6	5

Using the Mann–Whitney U test at the 5% significance level, determine whether customer satisfaction differs between the banks.

Task 3.
При измерении порога чувствительности к антибиотикам получены следующие величины:

Мужчины	39	36	31	35	29	34	38	
Женщины	32	30	28	30	33	37	28	27

Отличаются ли между собой пороги чувствительности у мужчин и женщин?

Task 4.
Даны две независимые выборки: определения уровня удовлетворённости жизнью. Группа А – мужчин и группа В - женщин. Проверить различаются ли между собой уровни удовлетворённости у мужчин и женщин? Уровень значимости $\alpha = 0.05$.

Группа А	15	18	22	21	19	17	20
Группа В	12	14	16	13	11	10	

Task 5.
При измерении порога чувствительности к витамину D получены следующие величины:

Группа A	5	7	7	9	10	11	12	8
Группа B	6	7	8	9	5	4	10	

Отличаются ли между собой пороги чувствительности мужчин и женщин?

Part II. Wilxocon test.

Task 6. Evaluating the Effect of a Training Program.

A company introduces a training program to improve employee productivity. The productivity scores of eight employees are measured before and after the program.

Employee	1	2	3	4	5	6	7	8
Before	55	62	58	70	64	59	66	61
After	60	65	63	72	68	61	70	66

At the 5% significance level, test whether the training program is associated with a change in productivity.

Task 7. Comparing Blood Pressure Before and After an Intervention.
A researcher investigates whether a lifestyle intervention changes systolic blood pressure. Blood pressure is measured in the same seven participants before and after the intervention.

Participant	1	2	3	4	5	6	7
Before (mmHg)	150	142	155	148	160	145	152
After (mmHg)	145	140	150	149	154	142	146

Using the Wilcoxon signed-rank test at the 5% significance level, determine whether blood pressure changed after the intervention.